Spatially-Correlated Distributionally Robust Scheduling for Carbon-Aware Data Centers

Phase 1 Results — The Spatial Question, Resolved

Marco Ortiz Togashi
Supervisor: Prof. Bissan Ghaddar
IE School of Science & Technology — Capstone in Operations Research

June 2026

Abstract

This chapter closes the central Phase 1 question: does modelling the *spatial correlation* of carbon intensity across electrically coupled regions improve a distributionally robust (DRO) data-center schedule? Extending the methodology of the status report to **three** US/Canada cases that span the full correlation spectrum, we find a **replicated, robustness-checked null**: the joint spatial covariance never beats an independent (shuffled-marginals) baseline by a material margin. The result is neither an artifact of weak correlation — we first *validate* strong, de-seasonalized correlation in two interconnections — nor an assertion: a **mean-ablation experiment** shows the covariance signal is genuine and exploitable but *dominated* by the mean carbon field, and a **tail-dependence** diagnostic shows the residual structure is non-elliptical (upper-tail-independent, radially asymmetric) and therefore invisible to a covariance-based ambiguity set by construction. Phase 1 thus delivers a *specification result* that directly motivates the Phase 2 copula extension.

1 Headline

> *Spatial correlation of carbon intensity does not translate into robust scheduling value: across the full correlation spectrum, in a Mahalanobis–Wasserstein DRO, modelling the joint spatial distribution buys essentially nothing over treating each region's carbon independently.*

Three findings make this a substantive, defensible result rather than a bare negative:

1. **It is not absence of correlation.** We validate strong, residual-surviving correlation in two interconnections (§2); the DRO's spatial assumption *is valid*. Validity of the assumption $\neq$ value from it.
2. **It is not an edge case.** The null holds in the strongly-correlated cases, in an adversarial *heterogeneous* case engineered to be diversifiable, and through a full robustness battery with multiple-testing correction (§3–§4).
3. **We prove *why*, causally.** A mean-ablation experiment (§5) demonstrates the covariance is real but mean-dominated; a tail-dependence diagnostic (§6) shows the exploitable structure is non-elliptical.

2 The three cases: a correlation spectrum

We transport the *identical* machinery of the status report — Algorithm 2b Mahalanobis–Wasserstein second-order cone program (SOCP), three nested constraint regimes, $\mathrm{CVaR}_{0.95}$ metric, blocked 5-fold cross-validation to select the Wasserstein radius ε, and a 1000-resample bootstrap confidence interval on the spatial gap — to three new region sets chosen to span the dependence spectrum (Table 1). All panels are hourly Electricity Maps lifecycle carbon intensity, 2021–2025 (43,824 hourly observations per zone).

Case	**Zones**	**Residual correlation**	**Mechanism**
`us_west`	CISO, BANC, LDWP, NEVP, AZPS	strong (0.72–0.78)	WECC: solar + weather
`taskc`	CA-ON, NYIS, MISO, PJM	strong (0.70–0.73)	East: weather fronts
`us_hetero`	CISO, ERCO, BPAT	near-zero (ERCO–BPAT 0.00)	solar / wind / hydro

Table 1: The three Phase 1 cases. `us_hetero` is adversarial by design: regions with different generation physics, chosen so their clean/dirty periods do *not* coincide — if spatial DRO value exists anywhere, it should appear here.

Correlation is genuine and survives de-seasonalization. Removing the per-zone hour-of-day mean (which collapses the spurious diurnal co-movement seen in the California and Iberia sub-grids of the status report), the two interconnected cases retain strong correlation: `us_west` CISO–NEVP 0.78, CISO–LDWP 0.72; `taskc` CA-ON–NYIS 0.73, MISO–PJM 0.70. This confirms the hypothesis that interconnected US/Canada grids carry real spatial carbon structure, unlike the European cases. Figure 1 (and `figures/ci_corr_heatmap_*`) shows the full correlation matrices.

3 The spatial gap: no robust value

The falsification test pairs the scheduler with either the *joint* sample covariance $\hat{\Sigma}$ or its *block-diagonal* (cross-region-zeroed) counterpart $\hat{\Sigma}^{\mathrm{shuf}}$; the out-of-sample $\mathrm{CVaR}_{0.95}$ gap shuf − joint measures the value of the spatial blocks (positive = joint better). The variable-capacity ceiling in regime R2 uses bounds $(x_{\min}, x_{\max}) = (50, 75)\,\mathrm{MW}$, calibrated to a loosely-binding regime on *training* data only (bind fraction ≈ 0.28–0.37). Results in Table 2.

Case	$\varepsilon^\star$ (CV)	**gap range (% of** CVaR**)**	**detectable**	**verdict**
`us_west`	1 (8/9)	$-0.005 \ldots +0.043$	1/9 (trivial, $+3\times10^{-6}$%)	null
`taskc`	1 (9/9)	$-0.022 \ldots +0.038$	4/9, *signs flip* across α	null (artifacts)
`us_hetero`	1 (9/9)	$-0.233 \ldots +0.062$	5/9, material ones *negative*	null (joint *hurts*)

Table 2: Spatial gap shuf − joint in $\mathrm{CVaR}_{0.95}$, 9 cells per case (3 regimes × 3 risk levels α). Every gap is a small fraction of one percent. CV selects $\varepsilon^\star = 1$ throughout, so the DRO is genuinely engaged — this is not a "model switched itself off" null. In `us_hetero` the only detectable cells are *negative*: estimating cross-region covariance that carries no exploitable signal merely adds estimation noise.

4 Reliability: robustness battery, multiple testing, temporal stability

Each case was re-run with seasonal-residual, AR(1)-residual, and Ledoit–Wolf-shrinkage covariance estimators. The largest |gap| under *any* estimator is 0.07% (`us_west`), 0.18% (`taskc`), 0.36% (`us_hetero`) of CVaR — all economically negligible. No cell shows a *material* beneficial effect that is sign-stable across the seasonal *and* AR(1) estimators; the cells that flip positive disagree by orders of magnitude across estimators (e.g. `taskc` R3/α0.30: seasonal -0.033% vs. AR(1) $+0.179\%$), the signature of ε-selection noise.

Multiple testing. Testing 144 non-ablation cells, some bootstrap CIs exclude zero by chance. Applying a Benjamini–Hochberg correction at $q = 0.05$ over all cells (bootstrap p-values), the largest *positive* gap that survives is 0.179% — precisely the AR(1) cell already rejected by the agreement rule. Nothing material survives both criteria.

Temporal stability. A walk-forward refit (train 2021–2023, test 2024) reproduces the null: max |gap| of 0.028%, 0.036%, 0.217% across the three cases. The result is not a 2025 artifact.

5 Mean-ablation: the causal mechanism

The preceding sections establish *that* the covariance adds no value; this experiment establishes *why*. We neutralize the mean field $\bar{\rho}$ used for *scheduling* while keeping *evaluation* on the real carbon panel. Two levels: LEVEL equalizes each region's time-average to the global mean (keeping the hour-of-day shape); FLAT sets a constant mean everywhere, so $\langle \bar{\rho}, x \rangle$ is constant under fixed demand and the schedule is driven *purely* by the $\varepsilon \| L^\top x \|_2$ covariance penalty — a *covariance-only world*. Results in Table 3.

Case	**baseline**	LEVEL	FLAT (covariance-only)
`us_west`	~ 0, none det.	~ 0, none	$\varepsilon^\star \to 0$; $-0.04 \ldots +0.004$
`taskc`	~ 0, sign-flips	~ 0	**$+0.14 \ldots +0.41$, all 9 det.**
`us_hetero`	negative	negative	**$+0.39 \ldots +1.46$, all 9 det.**

Table 3: Mean-ablation. Equalizing the *level* changes nothing — the diurnal mean shape still dominates. In the *covariance-only* world the joint covariance produces a material, fully-detectable advantage over the shuffled one, and CV no longer collapses to a single $\varepsilon^\star$. The spatial covariance therefore *does* carry genuine, exploitable information — it is simply *swamped* by the mean in the real problem. Identical gaps on a 2× denser ε grid confirm these are not grid artifacts.

This converts the null from an assertion into a demonstrated *ordering*: *the mean field dominates the schedule so completely that a real and exploitable second-moment signal contributes negligibly.* The cost confirms it is not free lunch: the FLAT schedule, by discarding the real mean, *worsens* absolute CVaR by +1.9–4.1% (`us_west`) and +1.6–3.8% (`us_hetero`) — the covariance-only gain ($\leq 1.46\%$) never repays it. `us_west` is the instructive exception ($\varepsilon^\star \to 0$, gap ~ 0): its strong *common-mode* positive correlation offers little even in isolation, because diversification needs heterogeneity, not co-movement.

6 Tail dependence: the structure the covariance cannot see

Why is even the exploitable-in-isolation covariance the *wrong* object? Because the genuine dependence lives in the tails and is non-elliptical. We estimate the empirical upper/lower tail-dependence coefficients χ_{U} (both regions *dirty* together — the CVaR-relevant co-movement) and χ_{L} (both *clean* together), and compare χ_{U} against the Gaussian-copula value at the matched correlation. A Gaussian copula — anything a covariance ball can represent — has *zero* asymptotic tail dependence, so an empirical excess above it would be structure the DRO misses.

Case	**Pair**	ρ	$\chi_{\mathrm{U}}^{\mathrm{emp}}$	$\chi_{\mathrm{U}}^{\mathrm{Gauss}}$	**excess**	$\chi_{\mathrm{L}}^{\mathrm{emp}}$
`us_west`	CISO | LDWP	0.72	0.25	0.41	−0.16	0.48
`us_west`	BANC | LDWP	0.65	0.17	0.35	−0.18	0.40
`taskc`	CA-ON | NYIS	0.73	0.38	0.42	−0.04	0.31
`us_hetero`	ERCO | BPAT	0.00	0.04	0.05	−0.02	0.02

Table 4: Residual tail dependence at $q = 0.95$. Two findings: (i) the empirical χ_{U} sits *at or below* the Gaussian benchmark (excess ≤ 0) — in the dirtiest hours regions *de-couple* (upper-tail independence), so there is nothing in the dirty tail for a richer model to exploit; (ii) $\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$ — regions go *clean* together more than *dirty* together. A Gaussian/elliptical copula forces $\chi_{\mathrm{L}} = \chi_{\mathrm{U}}$, so this radial asymmetry is non-elliptical structure a Mahalanobis–Wasserstein ball cannot encode.

7 Synthesis: why this is the expected outcome

1. **The mean field dominates** (proven in §5). The schedule is set by $\bar{\rho}$, shared identically by the joint and shuffled models; the covariance enters only as a second-order penalty.
2. **Positive correlation is common, non-diversifiable risk.** Diversification value needs regions that hedge each other; real grids co-move, and the genuinely clean escape (Ontario; hydro BPAT) is already identified by its low *mean*.
3. **Even zero correlation does not help** (`us_hetero`): with no cross-structure, fitting the joint covariance only adds estimation noise.
4. **The exploitable structure is non-elliptical** (§6) — invisible to a covariance/Wasserstein ambiguity set by construction.

8 Practical recommendation

For carbon-aware data-center scheduling under a covariance-based robust model, **a per-region marginal scheduler captures essentially all the value**. Modelling cross-region carbon dependence adds estimation burden and, in the heterogeneous case, measurable noise, for no robust benefit. Spatial dependence modelling should be reserved for (i) decision structures that can *act* on it (inter-region workload transfer) and (ii) dependence models that can represent its actual non-elliptical shape — both Phase 2 directions.

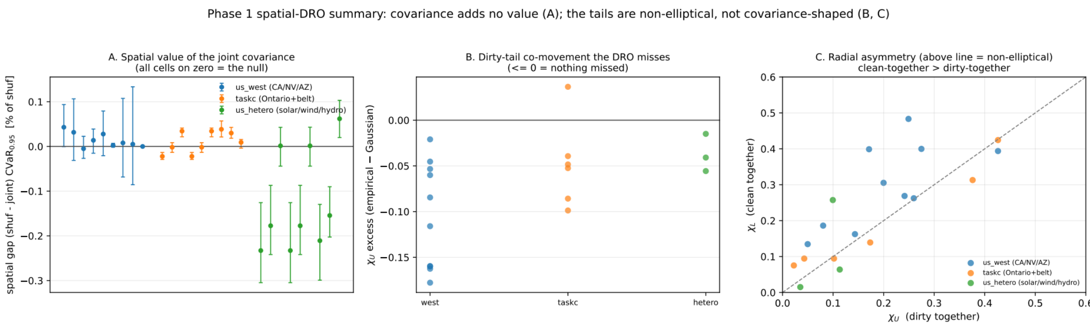

Figure 1: Phase 1 on one page. **(A)** spatial gap ± bootstrap CI for every case/regime/α cell, clustered on zero — the null. **(B)** upper-tail dependence excess (empirical − Gaussian) ≤ 0 — no dirty-tail structure is missed. **(C)** χ_{L} vs. χ_{U}: points above the diagonal (clean-together > dirty-together) mark the non-elliptical asymmetry. Regenerate with `python -m scripts.plot_case_summary`.

9 Conclusion and Phase 2 outlook

Phase 1 is a *specification result*: the second moment of spatial carbon dependence is both *dominated* (by the mean field) and the *wrong object* (the dependence is non-elliptical — upper-tail-independent and radially asymmetric). This is not a dead end but a precise mandate for Phase 2: model the **copula**, not the covariance. A vine-copula ambiguity set can represent asymmetric tail dependence ($\chi_{\mathrm{L}} \neq \chi_{\mathrm{U}}$, Clayton-type clean-together coupling) that the Mahalanobis–Wasserstein ground metric cannot express; in parallel, an inter-region *transfer* channel would give the decision a spatial degree of freedom on which the joint distribution could finally act. Phase 1 has both falsified the simple spatial-covariance hypothesis and identified, with evidence, exactly where the remaining value must lie.

Reproducibility. All results derive from `scripts/run_case_experiment.py` (region sets `us_west`, `taskc`, `us_hetero`; flags `--residualize`, `--shrinkage`, `--ablate-mean`, `--test-year`, `--eps-grid`), `scripts/calibrate_capacity.py`, `scripts/bh_correction.py`, and the diagnostic plotters. Summary tables for every number cited here are archived in `docs/results_snapshots/`.